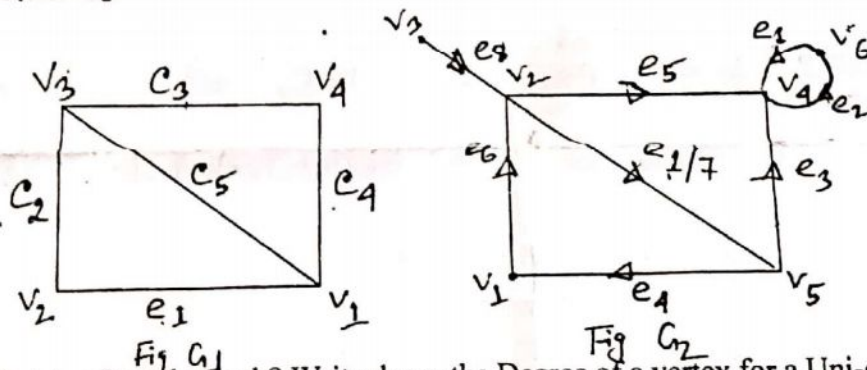


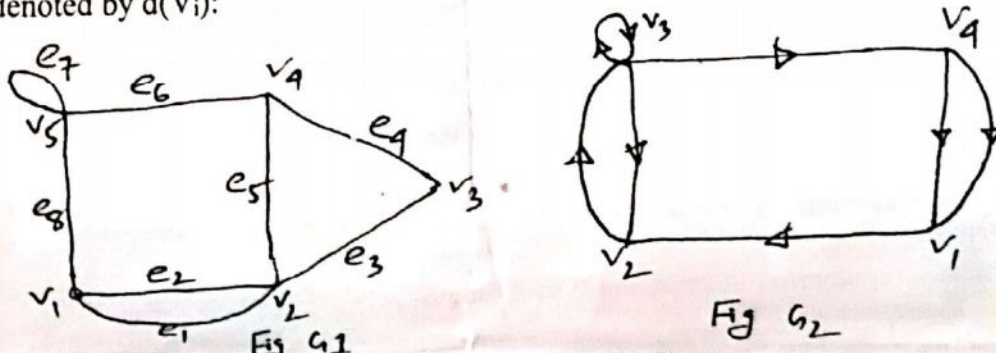
- ✓(a) Define Set with example 2
 (b) Explain Venn Diagram for Union and Intersection operations where, A and B are two sets. 6
 (c) Write down the Laws of set theory. 3
 (d) Prove that $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$,
 where A_1 and A_2 be sets with cardinalities $|A_1|$ and $|A_2|$. 3

- ✓(a) What is the Recursion? Obtain the recurrence relation for $G(k) = 2.4^k - 5(-3)^k$ 5
 (b) What is a Propositional Calculus? Find the direct and indirect proof of $p \rightarrow (q \rightarrow s)$,
 $\neg r \vee p \quad q \Rightarrow r \rightarrow s$ 6
 (c) Draw the Block Diagram with Truth Table for two input Logic Gate AND, OR and NOT 3

- ✓(a) What is an Incidence Matrix? Construct the Incidence matrix for the following graphs G_1 and G_2 . 6



- (b) What is a Pseudograph? Write down the Degree of a vertex for a Uni-directed graph G_1 and Directed graph G_2 respectively from the following figure, where the degree of a vertex denoted by $d(V_i)$: 6



- (c) Draw the digraph from the following adjacency matrix (AG) 2

$$AG = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

4. (a) Explain Euler paths and Euler circuits in Graphs with example. 4
- (b) What is meant by Functions and Inverse Functions with proper example? 3
- (c) Let, R be the relation with ordered pairs (Abdul, 22), (Kalam, 24), (Azad, 21), (Desire, 22), (Eddie, 24), and (Felicia, 22). Here each pair consists of a graduate student and this student's age. What is the function that this relation determines? 2
- (d) What are the differences among Injections, Surjections and Bijections? 3
- (e) Let, f and g be the functions from the set of integers to the set of integers defined by $f(x) = 6x + 3$ and $g(x) = 5x + 2$. What is the composition of f and g ? What is the composition of g and f ? 2
5. (a) Let, f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse? 2
- (b) Define binary relation with example. 2
- (c) Let, set $A = \{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$. 2
- (d) Explain reflexive relation and Equivalence Relation with suitable example. 4
- (e) What is the composite of the relations R and S where R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$? 2
- (f) What do you mean by Graphs of relation? 2
6. (a) Let, $A = \{1, 2, 3\}$ and R be the relation, $R = \{(1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$. Draw the digraph by using this relation. 2
- (b) Draw the Hasse diagram representing the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on set $\{1, 2, 3, 4, 6, 8, 12\}$. 3
- (c) Let, set $A = \{1, 2, 3\}$ and the relation r from A into A as $r = \{(1, 2), (1, 3), (2, 1), (1, 1), (2, 3)\}$. Draw the relation matrix. 2
- (d) Write short notes on Relational Database. 3
- (e) State and prove the Principle of Inclusion and Exclusion. 4
7. (a) Define semi groups and Monoids with example. 4
- (b) Write down the properties of a Group G , where G is a non empty set together with a binary operation $*$. 4
- (c) What is a Subring? Write down the relationship between a Commutative Ring and Zero Divisors. 4
- (d) Prove that for any two ideals A and B of a Ring R , product AB is an ideal of R . 2